

# Natural Units in Theoretical Physics: A Treatise in the Context of T0 Theory

## Contents

### Abstract

The use of natural units in theoretical physics is a fundamental concept that can be comprehensively explained and contextualized within the framework of T0 theory. This treatise illuminates the principle of dimensional reduction, the advantages for calculations, the particular relevance for T0 theory, and the necessity of explicit SI units in practice. Finally, it emphasizes the deeper insight that physics ultimately rests on dimensionless geometric relationships.

## .1 Basic Principle of Natural Units

### The Principle of Dimensional Reduction

In natural units, one sets fundamental constants to 1:

- **Speed of light:**  $c = 1$
- **Reduced Planck constant:**  $\hbar = 1$
- **Boltzmann constant:**  $k_B = 1$
- **Sometimes:**  $G = 1$  (Planck units)

### Mathematical Consequence

This does not mean that these constants "disappear," but that they serve as **scale setters**:

$$E = mc^2 \quad \Rightarrow \quad E = m \quad (\text{since } c = 1) \quad (1)$$

$$E = \hbar\omega \quad \Rightarrow \quad E = \omega \quad (\text{since } \hbar = 1) \quad (2)$$

## .2 Advantages for Calculations

### Simplified Formulas

**With SI units:**

$$E = \sqrt{(pc)^2 + (mc^2)^2} \quad (3)$$

**In natural units:**

$$E = \sqrt{p^2 + m^2} \quad (4)$$

### Transparent Dimensional Analysis

All quantities can be traced back to one fundamental dimension (typically energy):

| Quantity | Natural Dimension | SI Equivalent |
|----------|-------------------|---------------|
| Length   | $[E]^{-1}$        | $\hbar c/E$   |
| Time     | $[E]^{-1}$        | $\hbar/E$     |
| Mass     | $[E]$             | $E/c^2$       |

**Table 1:** Dimensional relationships in natural units

### .3 Particular Relevance in T0 Theory

#### Geometric Nature of Constants

T0 theory shows particularly clearly why natural units are fundamental:

$$\alpha = \xi \cdot \left( \frac{E_0}{1 \text{ MeV}} \right)^2 \quad (5)$$

This makes explicit that the fine structure constant is a **purely dimensionless geometric relationship**.

#### The $\xi$ -Parameter as Fundamental Geometry Factor

The derivation:

$$\xi = \frac{4}{3} \times 10^{-4} \quad (6)$$

is intrinsically dimensionless and represents the fundamental space geometry – independent of human units of measurement.

**Important:**  $\xi$  alone is not directly equal to  $1/m_e$  or  $1/E$ , but requires specific scaling factors for different physical quantities.

### .4 Derivation of the Fundamental Scaling Factor $S_{T0}$

#### The Fundamental Prediction of T0 Theory

T0 theory makes a remarkable prediction: the electron mass in geometric units is exactly:

$$m_e^{T0} = 0.511 \quad (7)$$

This is not a convention, but a **derived consequence** of the fractal space geometry via the  $\xi$  parameter.

#### Explicit Demonstration: Derivation vs. Reverse Calculation

Let us demonstrate explicitly that the scaling factor is derived, not reverse-calculated:

$$1. \text{ T0 derivation: } m_e^{T0} = 0.511 \quad (\text{from } \xi \text{ geometry}) \quad (8)$$

$$2. \text{ Experimental: } m_e^{SI} = 9.109 \times 10^{-31} \text{ kg} \quad (9)$$

$$3. \text{ T0 prediction: } S_{T0} = m_e^{SI}/m_e^{T0} = 1.7827 \times 10^{-30} \quad (10)$$

$$4. \text{ Empirical fact: } 1 \text{ MeV}/c^2 = 1.7827 \times 10^{-30} \text{ kg} \quad (11)$$

$$5. \text{ Conclusion: T0 theory predicts the MeV mass scale} \quad (12)$$

## Why This Is Not Circular Reasoning

Some might mistakenly think: "You're just defining  $S_{T0}$  to match  $1 \text{ MeV}/c^2$ ."

This misunderstands the logical flow:

- **Wrong interpretation (reverse calculation):**  $m_e^{T0} = \frac{m_e^{SI}}{1 \text{ MeV}/c^2}$  (circular)
- **Correct interpretation (derivation):**  $S_{T0} = \frac{m_e^{SI}}{m_e^{T0}}$  and this **happens to equal**  $1 \text{ MeV}/c^2$

The equality  $S_{T0} = 1 \text{ MeV}/c^2$  is a **prediction**, not a definition.

## Side-by-Side Comparison

| Conventional Physics                                                             | T0 Theory                                                  |
|----------------------------------------------------------------------------------|------------------------------------------------------------|
| $1 \text{ MeV}/c^2 = 1.782662 \times 10^{-30} \text{ kg}$ (arbitrary definition) | $m_e^{T0} = 0.511$ (derived from $\xi$ geometry)           |
| $m_e = 0.511 \text{ MeV}/c^2$ (independent measurement)                          | $S_{T0} = \frac{m_e^{SI}}{m_e^{T0}}$ (fundamental scaling) |
| Two independent facts                                                            | One <b>predicts</b> the other                              |

**Table 2:** Comparison of conventional vs. T0 interpretation of mass scales

The remarkable fact is: **Both approaches yield identical numbers, but T0 explains why.**

## The Coincidence That Isn't

What appears as a mere numerical coincidence is actually a fundamental prediction:

$$\text{T0 prediction: } S_{T0} = \frac{m_e^{SI}}{m_e^{T0}} = \frac{9.1093837 \times 10^{-31}}{0.511} \quad (13)$$

$$\text{Conventional definition: } 1 \text{ MeV}/c^2 = 1.782662 \times 10^{-30} \text{ kg} \quad (14)$$

These are **identical** not by definition, but because T0 theory correctly predicts the fundamental mass scale.

## The Profound Implication

**T0 theory does not "use" the MeV definition.  
It derives why the MeV has the mass scale it does.**

The conventional definition  $1 \text{ MeV}/c^2 = 1.782662 \times 10^{-30} \text{ kg}$  appears arbitrary, but T0 theory reveals it to be a consequence of fundamental geometry.

## Independent Verification

We can verify this independently:

- **Without T0:**  $1 \text{ MeV}/c^2 = 1.782662 \times 10^{-30} \text{ kg}$  (apparently arbitrary convention)
- **With T0:**  $S_{T0} = 1.782662 \times 10^{-30}$  (fundamental scaling derived from geometry)

- **Agreement:** The identical numerical value confirms T0's predictive power

This is analogous to how  $c = 299,792,458$  m/s appears arbitrary until one understands relativity.

## .5 Quantized Mass Calculation in T0 Theory

### Fundamental Mass Quantization Principle

In T0 theory, particle masses are **quantized** and follow from the fundamental geometry parameter  $\xi$  through discrete scaling relationships:

$$m_i^{T0} = n_i \cdot Q_m^{T0} \cdot f_i(\xi) \quad (15)$$

where:

- $n_i \in \mathbb{N}$  - Quantum number (discrete)
- $Q_m^{T0}$  - Fundamental mass quantum in T0 units
- $f_i(\xi)$  - Particle-specific geometry function

### Electron Mass as Reference

The electron mass serves as the fundamental reference mass:

$$\xi_e = \frac{4}{3} \times 10^{-4} \times f_e(1, 0, 1/2) \quad (16)$$

$$m_e^{T0} = Q_m^{T0} \cdot \frac{\xi}{\xi_e} = 0.511 \quad (17)$$

### Complete Particle Mass Spectrum

For detailed derivations of all elementary particle masses within the T0 framework, including quarks, leptons, and gauge bosons, refer to the separate comprehensive treatment "Particle Masses in T0 Theory" which provides:

- Complete mass calculations for all Standard Model particles
- Derivation of mass quantization rules
- Explanation of generation patterns
- Comparison with experimental values
- Fractal correction procedures for precision matching

## .6 Important: Explicit SI Units are Necessary for...

### 1. Experimental Verification

Every measurement is performed in SI units:

- Particle masses in  $\text{MeV}/c^2$
- Cross sections in barn
- Magnetic moments in  $\mu_B$

## 2. Technological Applications

- Detector design (lengths in m, times in s)
- Accelerator technology (energies in eV)
- Medical physics (dosage measurements)

## 3. Interdisciplinary Communication

- Astrophysics (redshifts, Hubble constant)
- Materials science (lattice constants)
- Engineering

## .7 Concrete Conversion in T0 Theory

### Example: Electron Mass

In T0 geometric units:

$$m_e^{T0} = 0.511 \quad (\text{as pure geometric number derived from } \xi) \quad (18)$$

In SI units:

$$m_e^{SI} = m_e^{T0} \cdot S_{T0} = 0.511 \cdot 1.782662 \times 10^{-30} = 9.1093837 \times 10^{-31} \text{ kg} \quad (19)$$

### The Fundamental Scaling Relationship

The conversion from T0 geometric quantities to SI units is accomplished by:

$$[SI] = [T0] \times S_{T0} \quad (20)$$

where  $S_{T0} = 1.782662 \times 10^{-30}$  is the fundamental scaling factor **derived** in Section ??, not defined.

## .8 Correct Energy Scale for the Fine Structure Constant

The fundamental relationship for the fine structure constant requires a precise energy reference:

$$\alpha = \xi \cdot \left( \frac{E_0}{1 \text{ MeV}} \right)^2 \quad (21)$$

$$\text{with } E_0 = 7.400 \text{ MeV} \quad (\text{characteristic energy}) \quad (22)$$

This yields:

$$\alpha = 1.333333 \times 10^{-4} \cdot (7.400)^2 \quad (23)$$

$$= 1.333333 \times 10^{-4} \cdot 54.76 \quad (24)$$

$$= 7.300 \times 10^{-3} \quad (25)$$

$$\frac{1}{\alpha} = 137.00 \quad (26)$$

The slight deviation from the experimental value  $1/\alpha = 137.036$  is due to higher-order fractal corrections that are accounted for in the complete correction procedure.

## .9 Integration of Fractal Correction into Natural Units

The formulas in T0 theory fit in natural units without explicit fractal correction, because these units isolate the geometric essence of the theory. For exact conversions to SI units, however, fractal correction is essential to incorporate self-similar corrections of the vacuum geometry.

### Why Do the Formulas Fit in Natural Units Without Fractal Correction?

In natural units, physics is reduced to a geometric, dimensionless basis (cf. Section ??). The fundamental constants serve only as a scale, and the core formulas hold approximately without additional corrections because:

- **The  $\xi$ -parameter is intrinsically dimensionless:**  $\xi$  represents the pure geometry of the vacuum field and acts like a "universal scaling factor."
- **Approximate validity for rough calculations:** Many T0 formulas are exact in the geometric ideal form, without correction.
- **Example: Electron mass in natural units:**

$$m_e^{T0} = 0.511 \quad (\text{geometric number, without correction}) \quad (27)$$

This "fits" immediately because  $\xi$  sets the geometric scale.

### Why is Fractal Correction Necessary for Exact SI Conversions?

SI units are human conventions that "contaminate" the geometric purity of T0 theory. To achieve exact agreement with experiments, fractal correction must be **explicitly applied** because:

- **Fractal self-similarity breaks scale invariance**
- **Conversion requires explicit scaling**
- **Cosmological reference effects**

### Mathematical Specification of Fractal Correction

The fractal correction is explicitly defined as:

$$f_{\text{fractal}}(E_0) = \prod_{n=1}^{137} \left( 1 + \delta_n \cdot \xi \cdot \left( \frac{4}{3} \right)^{n-1} \right) \quad (28)$$

where  $\delta_n$  are dimensionless coefficients describing the fractal structure at each stage.

### Comparison: Approximation vs. Exactness

## .10 Important Conceptual Clarifications

When applying T0 theory, note these fundamental distinctions:

- **T0 quantities** are geometric and derived from  $\xi$  (e.g.,  $m_e^{T0} = 0.511$ )
- **SI quantities** are physical measurements (e.g.,  $m_e^{\text{SI}} = 9.1093837 \times 10^{-31} \text{ kg}$ )
- $S_{T0}$  is the fundamental scaling between these realms, **derived** not defined
- The energy reference for  $\alpha$  is exactly  $E_0 = 7.400 \text{ MeV}$  in the geometric idealization
- All mass scales are **discretely quantized** in both T0 and SI representations

| Aspect            | Without fractal correction (T0 units)                    | With fractal correction (for SI conversion)          |
|-------------------|----------------------------------------------------------|------------------------------------------------------|
| Accuracy          | Approximate (~ 98–99 %, geometrically ideal)             | Exact (to $10^{-6}$ , matches CODATA measurements)   |
| Example: $\alpha$ | $\alpha \approx \xi \cdot (E_0)^2 \approx 1/137$ (rough) | $\alpha = 1/137.03599 \dots$ (via 137 stages)        |
| Mass calculation  | $m_e^{T0} = 0.511$ (geometric)                           | $m_e^{SI} = 9.1093837 \times 10^{-31}$ kg (physical) |
| Energy scale      | $E_0 = 7.400$ MeV (ideal)                                | $E_0 = 7.400244$ MeV (renormalized)                  |
| Scaling factor    | $S_{T0} = 1.782662 \times 10^{-30}$ (fundamental)        | $S_{T0} \cdot R_f$ (renormalized)                    |
| Advantage         | Fast, transparent calculations                           | Testability with experiments                         |
| Disadvantage      | Ignores fractal subtleties                               | Complex (iteration over resonance stages)            |

**Table 3:** Comparison of geometric idealization in T0 units and physical exactness with fractal correction.

## .11 Special Significance for T0 Theory

### The Deeper Insight

T0 theory reveals that natural units are not merely a calculational convenience, but express the **true geometric nature of physics**:

- $\xi$  is the fundamental dimensionless geometry constant
- $S_{T0}$  connects geometric idealization to physical measurement
- **T0 quantities** represent the ideal geometric forms
- **SI quantities** are their measurable projections into our physical reality
- **Particle masses** are quantized geometric patterns in both realms

### Practical Implications

1. **Theoretical development:** Work in T0 units using geometric quantities
2. **Fundamental scaling:** Apply  $S_{T0}$  to project to physical reality
3. **Predictions:** Convert to SI units for experimental verification
4. **Verification:** Compare with measured SI values
5. **Quantization:** Respect the discrete nature of all physical scales

## .12 Notation and Symbols

## .13 Fundamental Relationships

## .14 Conversion Factors



| Symbol               | Meaning and Explanation                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| $c$                  | Speed of light in vacuum; fundamental constant of nature                               |
| $\hbar$              | Reduced Planck constant                                                                |
| $k_B$                | Boltzmann constant                                                                     |
| $G$                  | Gravitational constant                                                                 |
| $E$                  | Energy; in natural units dimensionally equivalent to mass and frequency                |
| $m$                  | Mass; in natural units $m = E$ (since $c = 1$ )                                        |
| $p$                  | Momentum; in natural units dimensionally equivalent to energy                          |
| $\omega$             | Angular frequency; in natural units $\omega = E$ (since $\hbar = 1$ )                  |
| $\alpha$             | Fine structure constant; dimensionless coupling constant                               |
| $\xi$                | Fundamental geometry parameter of T0 theory; $\xi = \frac{4}{3} \times 10^{-4}$        |
| $E_0$                | Reference energy in T0 theory; $E_0 = 7.400$ MeV                                       |
| $m_e^{\text{T0}}$    | Electron mass in T0 units; $m_e^{\text{T0}} = 0.511$ (geometric)                       |
| $m_e^{\text{SI}}$    | Electron mass in SI units; $m_e^{\text{SI}} = 9.1093837 \times 10^{-31}$ kg (physical) |
| $[E]$                | Energy dimension; fundamental dimension in natural units                               |
| SI                   | International System of Units (physical measurements)                                  |
| T0                   | T0 geometric units (ideal geometric forms)                                             |
| $S_{T0}$             | Fundamental scaling factor; $S_{T0} = 1.782662 \times 10^{-30}$                        |
| $R_f$                | Fractal correction factor                                                              |
| $f_{\text{fractal}}$ | Fractal correction function                                                            |
| $Q_m^{\text{T0}}$    | Fundamental mass quantum in T0 units                                                   |
| $Q_m^{\text{SI}}$    | Fundamental mass quantum in SI units                                                   |
| $n_i$                | Quantum number for particle $i$ ; $n_i \in \mathbb{N}$ (discrete)                      |
| $\delta_n$           | Fractal correction coefficients; dimensionless                                         |

**Table 4:** Explanation of the notation and symbols used

| Relationship                                                 | Meaning                                               |
|--------------------------------------------------------------|-------------------------------------------------------|
| $E = m$                                                      | Mass-energy equivalence (since $c = 1$ )              |
| $E = \omega$                                                 | Energy-frequency relationship (since $\hbar = 1$ )    |
| $[L] = [T] = [E]^{-1}$                                       | Length and time have same dimension as inverse energy |
| $[m] = [p] = [E]$                                            | Mass and momentum have same dimension as energy       |
| $\alpha = \xi(E_0/1\text{MeV})^2$                            | Fundamental relationship in T0 theory                 |
| $m_i^{\text{T0}} = n_i \cdot Q_m^{\text{T0}} \cdot f_i(\xi)$ | Quantized mass formula in T0 units                    |
| $m_i^{\text{SI}} = m_i^{\text{T0}} \cdot S_{T0}$             | Fundamental scaling to SI units                       |
| $S_{T0} = \frac{m_e^{\text{SI}}}{m_e^{\text{T0}}}$           | Definition of fundamental scaling factor              |

**Table 5:** Fundamental relationships in T0 theory and scaling to physical units

| Quantity     | Conversion Factor          | Value                                      |
|--------------|----------------------------|--------------------------------------------|
| $S_{T0}$     | Fundamental scaling factor | $1.782662 \times 10^{-30}$                 |
| $m_e^{T0}$   | Electron mass (T0 units)   | 0.511                                      |
| $m_e^{SI}$   | Electron mass (SI units)   | $9.1093837 \times 10^{-31}$ kg             |
| 1 MeV/ $c^2$ | Conventional mass unit     | $1.782662 \times 10^{-30}$ kg              |
| 1 MeV        | Energy in joules           | $1.602176 \times 10^{-13}$ J               |
| 1 fm         | Length in natural units    | $5.06773 \times 10^{-3}$ MeV <sup>-1</sup> |

**Table 6:** Fundamental conversion factors between T0 geometric units and SI physical units